

MD ABU SAYEB RAYHAN

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Research Interests

Robot perception, embodied AI and intelligent autonomous systems; computer vision (multi-object tracking, activity recognition, pose estimation), multimodal vision-language models and generative models for simulation and robotic learning.

Education

Bachelor of Science, Computer Science & Engineering GPA: 3.4 / 4.0 Jan 2019 – May 2024
Comilla University, Bangladesh

Relevant Coursework: Artificial Intelligence, Algorithm Design, Theory of Computation, Introduction to Data Science, Object-Oriented Programming, Research-Based Project.

Bachelor's Thesis: *Text-to-Image Generation using Diffusion Models* (Sep 2023 – Jan 2024). Fine-tuned Stable Diffusion (U-Net backbone) on a custom dataset of Bangladeshi landmarks (Salbon Bihar, Comilla University Gate). Investigated prompt engineering and conditioning mechanisms for generative fidelity; applied OpenCV for post-processing and artefact reduction.

Publications

- “**Forensic Invariant Learning for Synthetic Image Detection: Bridging Benford’s Law and Topological Analysis with Machine Learning Ensembles.**” Presented at **IEEE QPAIN 2026**; to appear in *IEEE Xplore* (Scopus-indexed).
- “**BD Eye-Doc: When Computer Vision Meets Conversational AI for Scalable Eye Disease Screening in Bangladesh.**” Presented at **IEEE QPAIN 2026**; to appear in *IEEE Xplore* (Scopus-indexed).
- “**Is CNN Dead? A Comprehensive Empirical Analysis of Convolutional Neural Networks, Ensemble Methods, and Vision Transformers for Rice Leaf Disease Classification in Low-Resource Agricultural Settings.**” Presented – **ECCT 2026**. To appear in *ACSR*, Atlantis Press (Springer Nature).

Research Experience

Research Assistant – Comilla University Supervisor: *Partha Chakraborty*, Associate Professor Jan 2023 – Mar 2024

- Contributed to empirical studies in Human–Robot Interaction (HRI) and Optical Character Recognition (OCR): designed datasets, conducted experiments and performed quantitative analysis.
- Prepared and validated results subsequently incorporated into peer-reviewed conference publications.

Professional Experience

Data Scientist – Othello Financial Service Co. Ltd., Japan Oct 2024 – Oct 2025

- Investigated multi-object tracking (ByteTrack), pose estimation (MoveNet) and human activity recognition (MMAction2) for automated retail behaviour analysis.
- Designed a spatial clustering pipeline (OPTICS on pose-estimation outputs) for group-dynamics modelling; parallel inference optimisation reduced processing latency by approximately 70%.
- Developed a video event-segmentation system using a fine-tuned YOLO-based detector (90%+ accuracy); automated highlight extraction cut NFT production cost by 50%.
- Implemented a RAG-based conversational system (LangChain, Pinecone, GPT-4) for context-aware document querying; deployed on Microsoft Azure.
- Integrated IoT edge devices for automated retail data acquisition feeding real-time computer vision pipelines.

AI Engineer – Bangladesh Government Project (Contractual) Feb 2024 – Jul 2024

- Developed a multilingual (Bangla/English) NLP + OCR pipeline for keyword and context-based document retrieval across PDF, Word, Excel and image formats; built as a desktop application (PyQt + ChromaDB).
- Coordinated implementation within a three-person team across full product lifecycle from design to deployment.

Technical Skills

- Programming:** Python (advanced), C/C++, SQL, Java
- ML / AI Frameworks:** PyTorch, TensorFlow, OpenCV, MMAction2, YOLO, ByteTrack, MoveNet, Stable Diffusion, LangChain, Hugging Face, Scikit-learn, Pandas
- Methods:** Object Detection, Action Recognition, Pose Estimation, Spatial Clustering, Diffusion Models, OCR, RAG, Dense Embeddings, Temporal Segmentation, NLP
- Infrastructure:** Microsoft Azure (Databricks, DevOps), Docker, CI/CD, Git; Flask, PyQt; Pinecone, ChromaDB, MongoDB, Azure SQL; Power BI

Honours & Certifications

- Certified, *Skill Development for Mobile Game & Application Development* – ICT Division, Government of Bangladesh (2022)
- Participant, Inter-University Programming Contest, Comilla University (2019–2021)
- Served as a Joint Secretary, North Bengal Student Association, Comilla University(2023-2024)

Language Proficiency

English – Fluent | Bangla – Native | Japanese – Basic Daily Conversation (JLPT N4 equivalent)

References

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